KING SALMAN INTERNATIONAL UNIVERSITY

Field of ENGINEERING

Engineering Program
All Intelligence Engineering Program

MAT121 Statics

3 Cr. Hrs. = (***2*** LCT + ***2*** TUT + ***0*** LAB + ***0*** OTH)

Lecture : (Equilibrium of Particle)

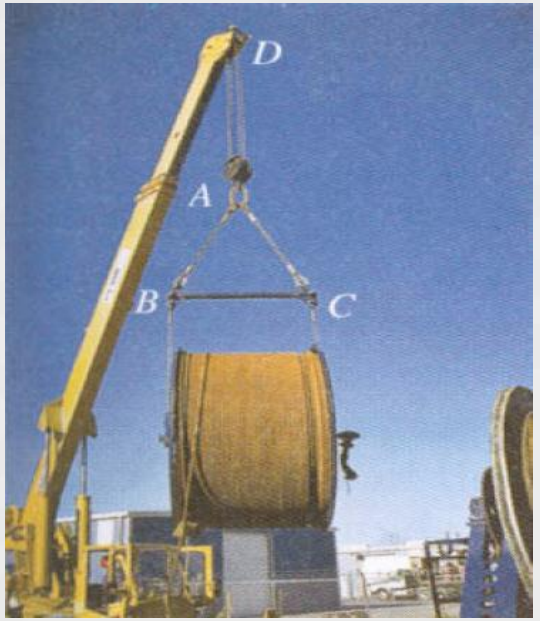
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Dr : (Dr. Haytham Essam)

Date : 21 / 10 / 2026

Equilibrium of a particle

Definition

When **the resultant of all the forces acting on a particle is zero**, the particle is in equilibrium

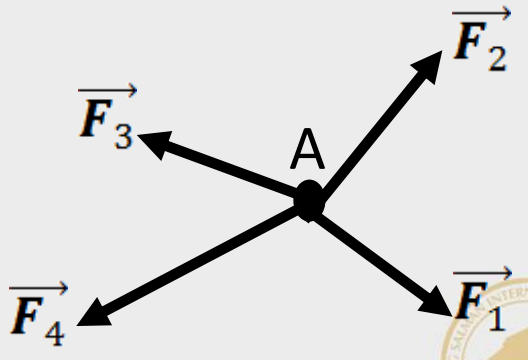


Condition of Equilibrium

Consider the four concurrent forces at equilibrium particle A as shown in Figure.

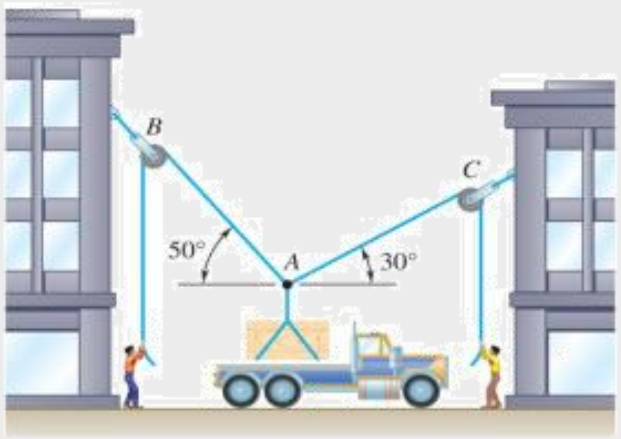
$$\vec{R} = \sum \vec{F} = 0$$

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 = 0$$

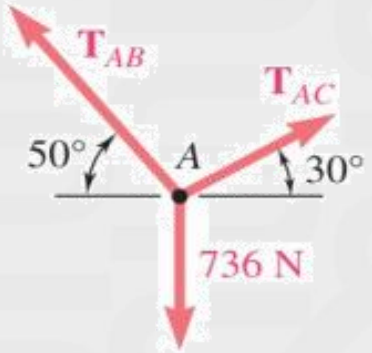


Procedure for solving problems

1- Draw the Free Body Diagram



Space diagram



Free-body diagram

A) Weight: w

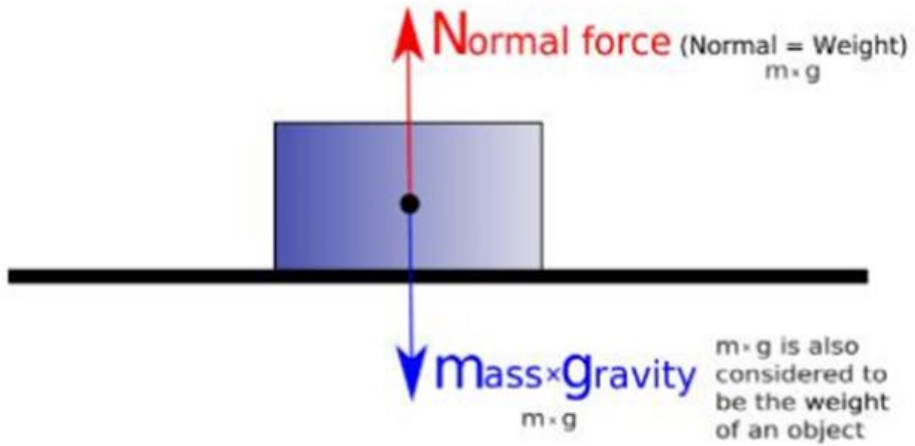


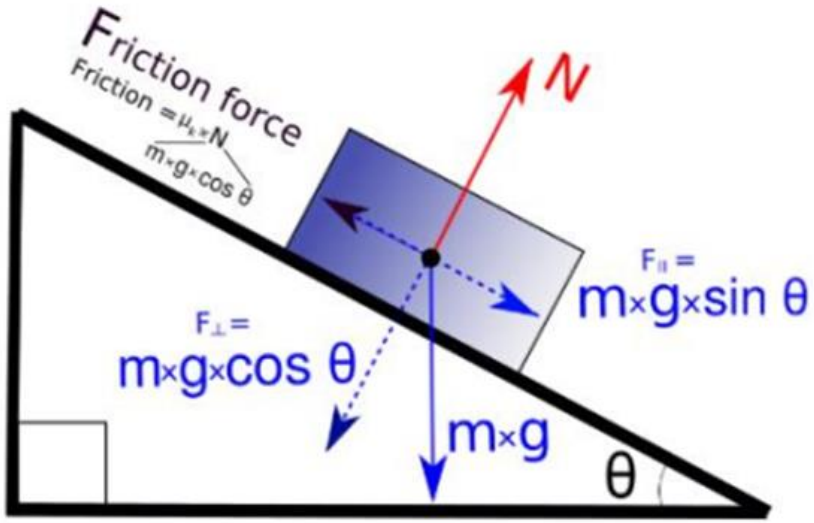
$$w = m * g$$

Procedure for solving problems

B)-Normal Force

It is a contact interaction force between surfaces. It always acts perpendicular to the surfaces and out of the surface.



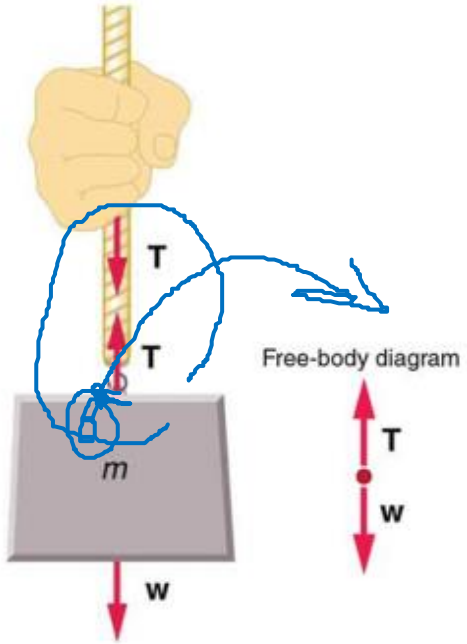


For a body resting on a given flat surface, then the normal force N is equal to the weight.

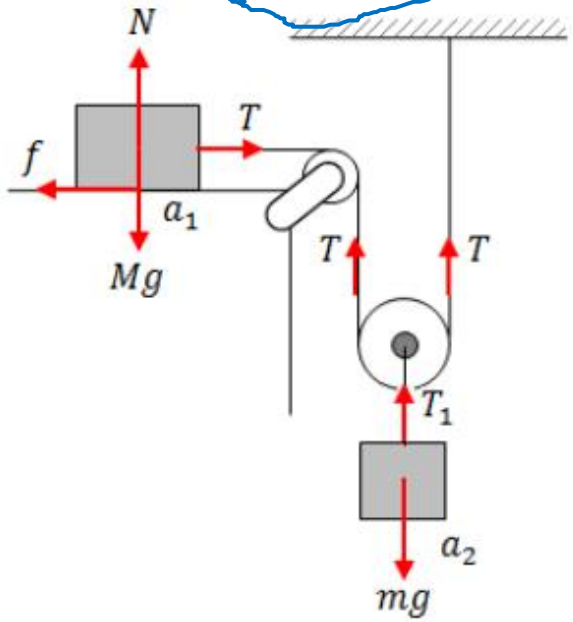
For a body resting on an inclined surface, then the normal force N is not equal to the weight.

Procedure for solving problems

C)-Tension in cables and pulleys







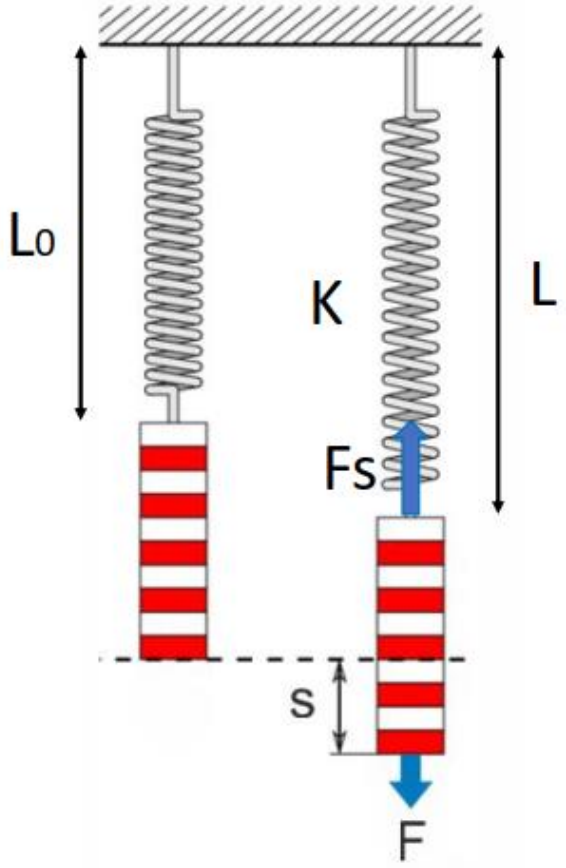
- Cables are weightless.
- Cable carry only tension forces directed along the axes of the cables.

- If the pully has friction T1 ≠ T2
- If the pully has frictionless T1 = T2

Procedure for solving problems

D)-Spring Force

Spring force comes from the displacement of molecules. It is always opposite the displacement of spring.



The spring force can be calculated using the following formula:

$$F_s = -K\,S$$

$$F_s = -K\,(L - L_0)$$

Fs = Spring Force [N]

K = spring stiffness [N/m]

L0 = Free, initial, unstretched length [m]

L = Final spring length [m]

S = Spring deflection or elongation [m]

The minus sign in the equation means that, relative to the resting position, the direction of deflection of spring is opposite to the spring force.

Equilibrium of a particle

2- All Forces in Cartesian form representation

3-Apply the equilibrium equation $\vec{R} = \sum \vec{F} = 0$

4- Equating coefficients

$i \longrightarrow \sum f_x = 0 \text{ ---------- } -1$

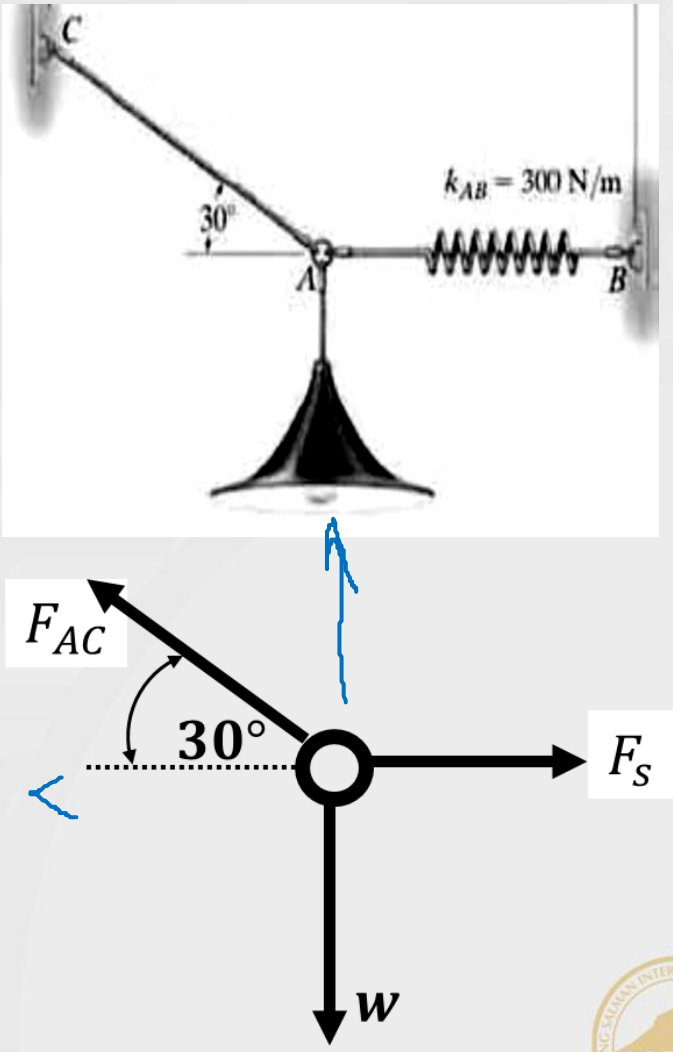
$j \longrightarrow \sum f_y = 0 \text{ ---------- } -2$

$k \longrightarrow \sum f_z = 0 \text{ ---------- } -3$ **3D**

5- Solve the resulting equations

Example 1

Determine the required length of spring lie in Fig. so that the 8-kg lamp can be suspended in the position shown. The unstreched length of the spring AB is $l_0 = 0.4\,m$, and the spring has a stiffness of $k = 300N/m$.



SOLUTION:

$\vec{F}_s = F_s\,\boldsymbol{i}$

$\vec{F}_{AC} = -F_{AC}\,cos\,30\,\boldsymbol{i} + F_{AC}\,sin\,30\,\boldsymbol{j}$

$\vec{F}_{AC} = -0.87F_{AC}\boldsymbol{i} + 0.5F_{AC}\boldsymbol{j}$

$\vec{w} = -8 \times 9.8\,\boldsymbol{j}$

$\vec{w} = -78.4\,\boldsymbol{j}$

Example 1

$$\vec{R} = \sum \vec{F} = \vec{F}_s + \vec{F}_{AC} + \vec{w} = 0$$

Equating coefficients of the unit vectors i and j

i ⟹ $F_s - 0.87F_{AC} = 0$----------(1)

j ⟹ $0.5\,F_{AC} - 78.4 = 0$----------(2)

From equations 1&2

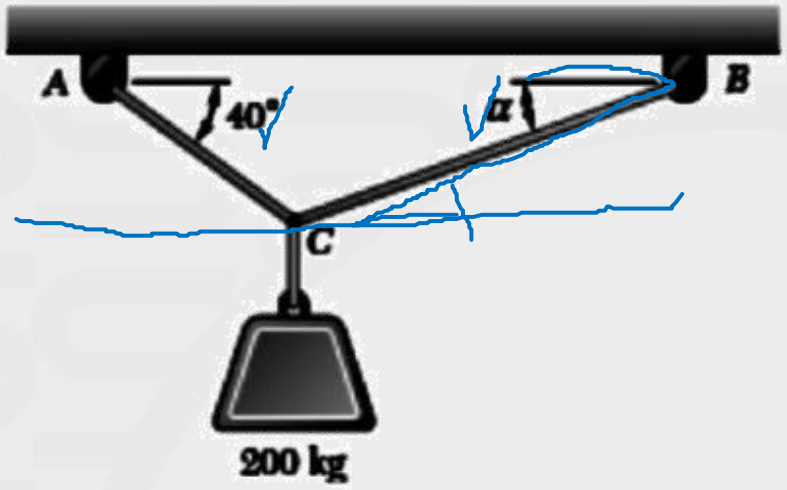
$$F_{AC} = 156.8\ \text{N} \qquad F_s = 135.8\ N$$

$$F_s = k(l - l_0)$$

$$135.8 = 300(l - 0.4) \Longrightarrow \underline{l = 0.85\ m}$$

Example 2

Two cables are tied together at C and are loaded as shown. Knowing that α = 20°, determine the tension (a) in cable AC, (b) in cable BC.



➢ **First draw free body diagram:**

- Isolate point of study C.
- Draw all external forces that affect point C, where there are:
  * Tension forces in cables CA and CB.
  * Weight due to hanging mass, where:

$$W = m \times g$$

$$W = 200 \times 9.8 = 1960\ N$$

Example 2

- **Second apply equilibrium equations**

$$\because \rightarrow_+ \sum F_x = 0$$

$$\therefore T_{CB} \cos 20 - T_{CA} \cos 40 = 0$$

$$\therefore T_{CB} = \frac{T_{CA} \cos 40}{\cos 20}$$

$$\therefore T_{CB} = 0.815\, T_{CA} \quad \Longrightarrow \quad I$$

$$\because \uparrow_+ \sum F_y = 0$$

$$\therefore T_{CB} \sin 20 + T_{CA} \sin 40 - W = 0$$

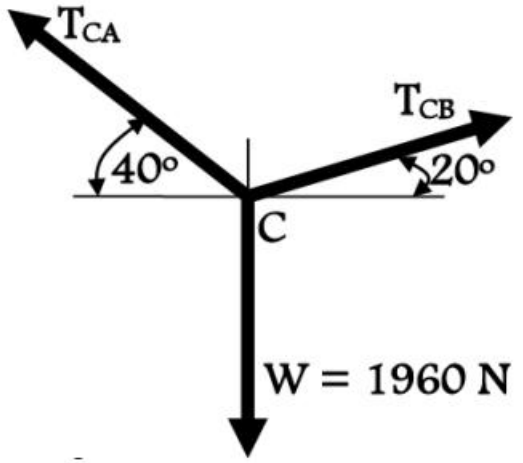
$$\therefore T_{CB} \sin 20 + T_{CA} \sin 40 - 1960 = 0$$

$$\therefore T_{CB} \sin 20 + T_{CA} \sin 40 = 1960 \quad \Rightarrow \quad II$$

By Substitution from I in II

$$\therefore T_{CA} = 2126.888\ N$$

$$\therefore T_{CB} = 1733.414\ N$$



Steps of Equilibrium 2D vs 3D

2D Particle Equilibrium

2 Dimensions



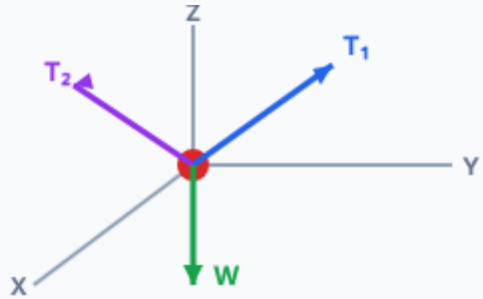
1. Draw Free Body Diagram (FBD)
2. Express all forces in Cartesian vector form
3. Apply equilibrium condition: $\sum \vec{F} = 0$
4. Equate scalar components to zero:

$$i \rightarrow \Sigma\ Fx = 0 \quad (1)$$
$$j \rightarrow \Sigma\ Fy = 0 \quad (2)$$

5. Solve the 2 resulting scalar equations

3D Particle Equilibrium

3 Dimensions



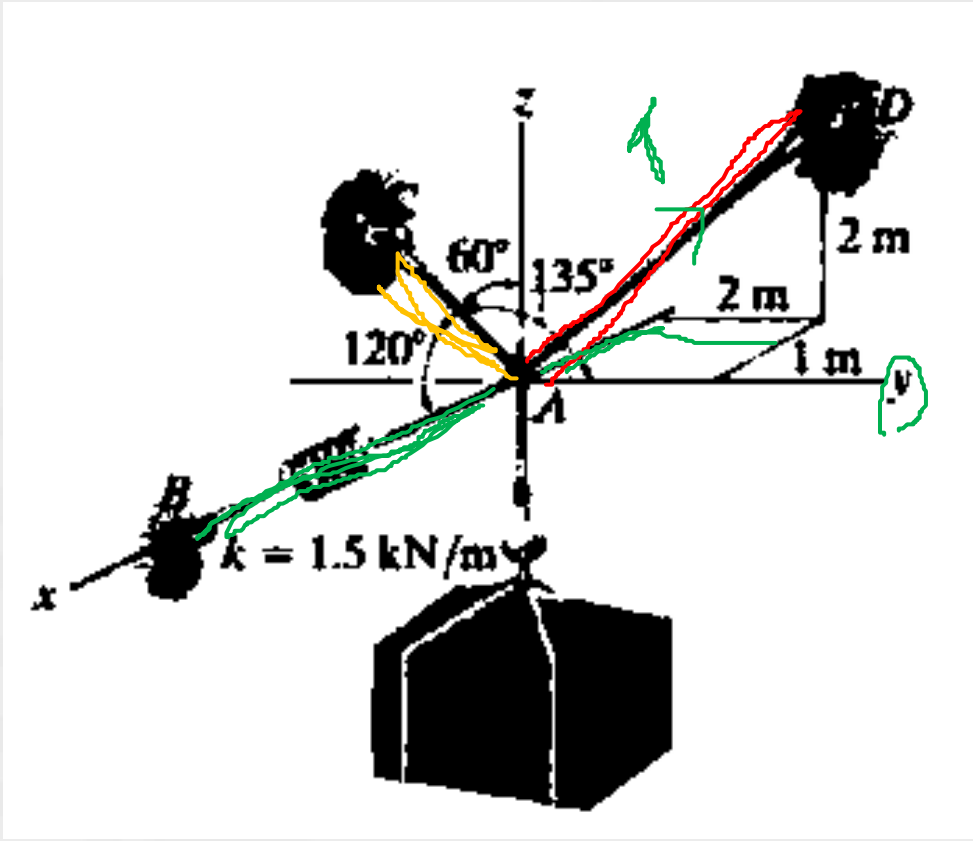
1. Draw Free Body Diagram (FBD)
2. Express all forces in Cartesian vector form
3. Apply equilibrium condition: $\sum \vec{F} = 0$
4. Equate scalar components to zero:

$$i \rightarrow \Sigma\ Fx = 0 \quad (1)$$
$$j \rightarrow \Sigma\ Fy = 0 \quad (2)$$
$$k \rightarrow \Sigma\ Fz = 0 \quad (3)$$

5. Solve the 3 resulting scalar equations

Example 3

Determine the tension in each cord used to support the 100-kg crate shown in Figure.



SOLUTION:

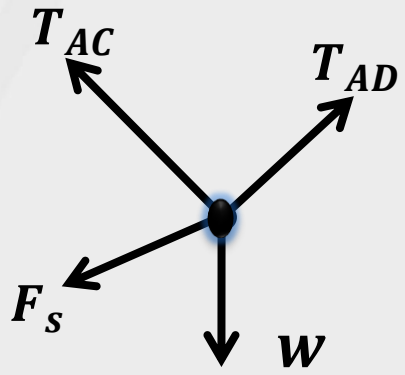
$$\vec{w} = -100 \times 9.81\, \boldsymbol{k}$$

$$\vec{w} = -981\, \boldsymbol{k}$$

$$\vec{\boldsymbol{F}}_{\boldsymbol{s}} = F_s\, \boldsymbol{i}$$

$$\vec{\boldsymbol{T}}_{\boldsymbol{AC}} = T_{AC} cos\ 120\, \boldsymbol{i} + T_{AC} cos\ 135\, \boldsymbol{j} + T_{AC} cos\ 60\, \boldsymbol{k}$$

$$\vec{\boldsymbol{T}}_{\boldsymbol{AC}} = -0.5 T_{AC} \boldsymbol{i} - 0.71 T_{AC} \boldsymbol{j} + 0.5 T_{AC} \boldsymbol{k}$$



Example 3

$$\vec{T}_{AD} = T_{AD}\left[\frac{-i+2j+2k}{\sqrt{1^2+2^2+2^2}}\right]$$

$$\vec{T}_{AD} = -0.33T_{AD}i + 0.67T_{AD}j + 0.67T_{AD}k$$

$$\vec{R} = \sum \vec{F} = \vec{F}_s + \vec{T}_{AC} + \vec{T}_{AD} + \vec{w} = 0$$

i ⟹ $F_s - 0.5T_{AC} - 0.33T_{AD} = 0$----------(1)

j ⟹ $0 - 0.71T_{AC} + 0.67T_{AD} = 0$----------(2)

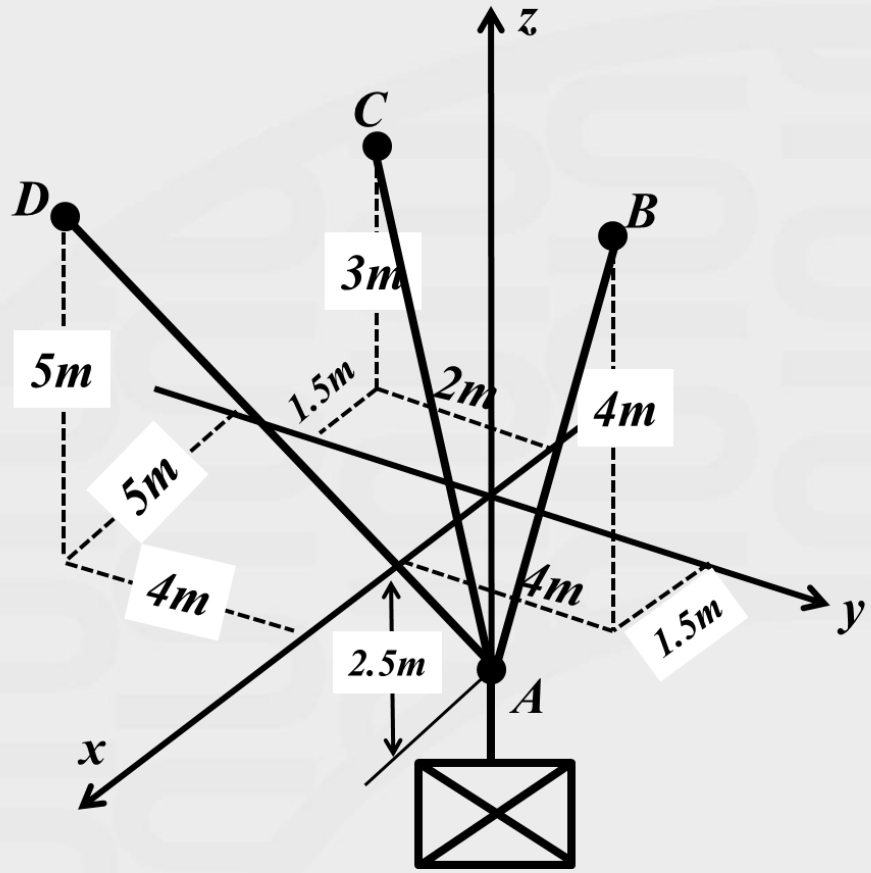
k ⟹ $0 + 0.5T_{AC} + 0.67T_{AD} = 981$----------(3)

From equations 1&2&3

$F_s = 694\ N,$ $T_{AC} = 813\ N$, $T_{AD} = 862\ N$

Example 4

Determine the tension in the cables in order to support the 1600-N crate in the equilibrium position shown.



SOLUTION:

$A = (0,0,-2.5)\ B = (1.5,4,4)$

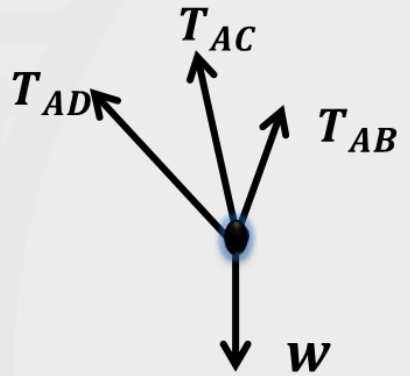
$C = (-1.5,-2,3)\ \ D = (5,-4,5)$

$$\vec{T}_{AB} = T_{AB}\left[\frac{1.5\,i+4\,j+6.5\,k}{\sqrt{1.5^2+4^2+6.5^2}}\right]$$

$$\vec{T}_{AB} = 0.193T_{AB}i + 0.514T_{AB}j + 0.836T_{AB}k$$

$$\vec{T}_{AC} = T_{AC}\left[\frac{-1.5\,i-2\,j+5.5\,k}{\sqrt{1.5^2+2^2+5.5^2}}\right]$$

$$\vec{T}_{AC} = -0.248T_{AC}i - 0.331T_{AC}j + 0.91T_{AC}k$$



Example 4

$$\vec{T}_{AD} = T_{AD}\left[\frac{5\,i - 4\,j + 7.5\,k}{\sqrt{5^2+4^2+7.5^2}}\right]$$

$$\vec{T}_{AD} = 0.51T_{AD}\,i - 0.41T_{AD}\,j + 0.76T_{AD}\,k$$

$$\vec{W} = -1600\,k$$

$$\vec{R} = \sum \vec{F} = \vec{T}_{AB} + \vec{T}_{AC} + \vec{T}_{AD} + \vec{W} = 0$$

$i \Rightarrow 0.193T_{AB} - 0.248T_{AC} + 0.51T_{AD} = 0$----------(1)

$j \Rightarrow 0.514T_{AB} - 0.331T_{AC} - 0.41T_{AD} = 0$----------(2)

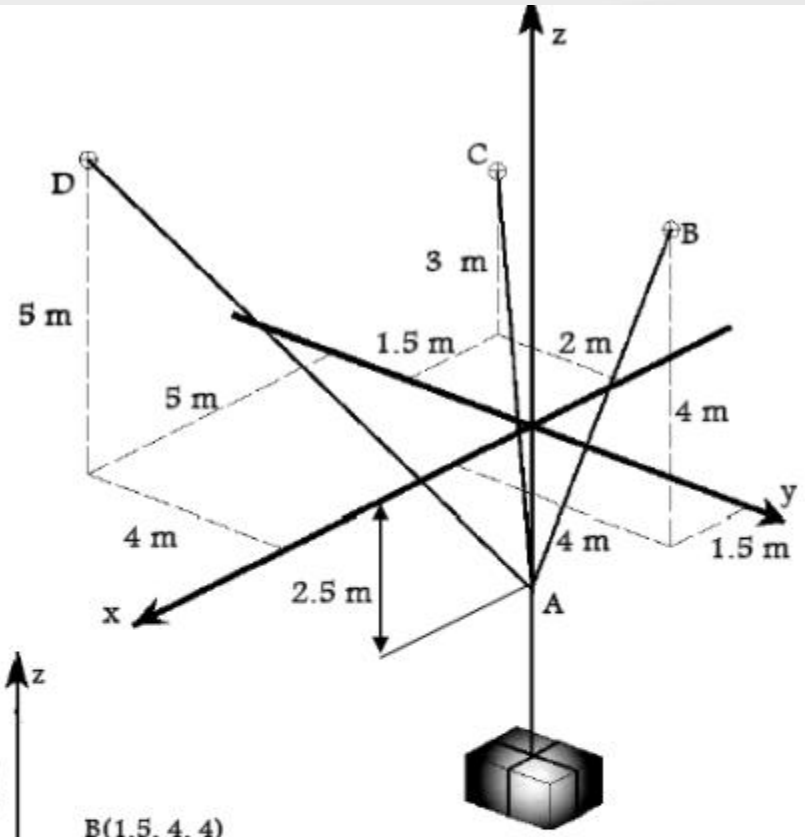
$k \Rightarrow 0.836T_{AB} + 0.91T_{AC} + 0.76T_{AD} = 1600$----------(3)

From equations 1&2&3

$T_{AB} = 739\ N$, $T_{AC} = 933\ N$, $T_{AD} = 174\ N$

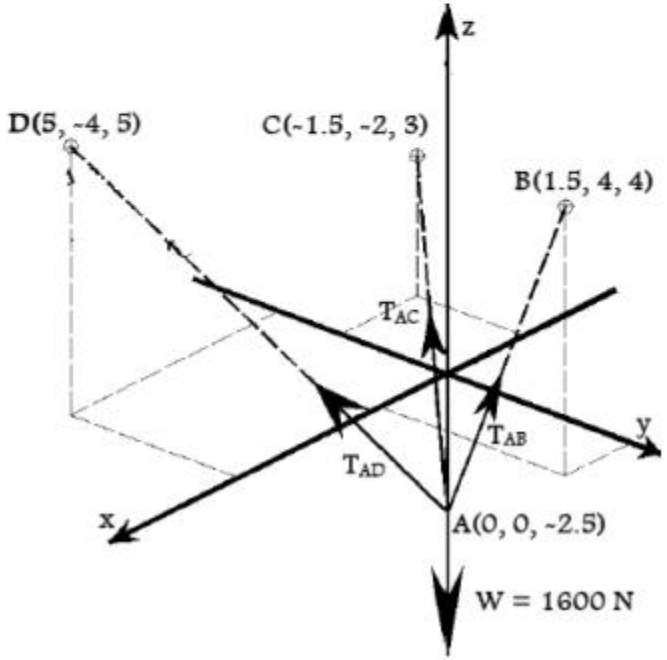
Example 5

A 1600 N crate is supported by three cables as shown. Determine the tension in each cable.



➢ **First draw free body diagram:**

- Isolate point of study A.
- Draw all external forces that affect point A, where there are:
  - Forces in cables AC, AB and AD.
  - Weight due to hanging crate.



Example 5

➢ **Second write each force in vector form:**

For weight:

$$\vec{W} = -1600\,\vec{k}$$

For tension force in cable AB:

$$\because \vec{T}_{AB} = T_{AB}\frac{(x_B - x_A)\vec{i} + (y_B - y_A)\vec{j} + (z_B - z_A)\vec{k}}{\sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}}$$

$$\therefore \vec{T}_{AB} = T_{AB}\frac{(1.5 - 0)\vec{i} + (4 - 0)\vec{j} + (4 + 2.5)\vec{k}}{\sqrt{(1.5)^2 + (4)^2 + (6.5)^2}}$$

$$\therefore \vec{T}_{AB} = 0.193T_{AB}\vec{i} + 0.514T_{AB}\vec{j} + 0.836T_{AB}\vec{k}$$

For tension force in cable AC:

$$\because \vec{T}_{AC} = T_{AC}\frac{(x_C - x_A)\vec{i} + (y_C - y_A)\vec{j} + (z_C - z_A)\vec{k}}{\sqrt{(x_C - x_A)^2 + (y_C - y_A)^2 + (z_C - z_A)^2}}$$

$$\therefore \vec{T}_{AC} = T_{AC}\frac{(-1.5 - 0)\vec{i} + (-2 - 0)\vec{j} + (3 + 2.5)\vec{k}}{\sqrt{(-1.5)^2 + (-2)^2 + (5.5)^2}}$$

$$\therefore \vec{T}_{AC} = -0.248T_{AC}\vec{i} - 0.331T_{AC}\vec{j} + 0.91T_{AC}\vec{k}$$

For tension force in cable AD:

$$\because \vec{T}_{AD} = T_{AD}\frac{(x_D - x_A)\vec{i} + (y_D - y_A)\vec{j} + (z_D - z_A)\vec{k}}{\sqrt{(x_D - x_A)^2 + (y_D - y_A)^2 + (z_D - z_A)^2}}$$

$$\therefore \vec{T}_{AD} = T_{AD}\frac{(5 - 0)\vec{i} + (-4 - 0)\vec{j} + (5 + 2.5)\vec{k}}{\sqrt{(5)^2 + (-4)^2 + (7.5)^2}}$$

$$\therefore \vec{T}_{AD} = 0.507T_{AD}\vec{i} - 0.406T_{AD}\vec{j} + 0.761T_{AD}\vec{k}$$

$$\because \sum F_x = 0 = 0.193T_{AB} - 0.248T_{AC} + 0.507T_{AD} \Rightarrow \quad I$$

$$\because \sum F_y = 0 = 0.514T_{AB} - 0.331T_{AC} - 0.406T_{AD} \Rightarrow \quad II$$

$$\because \sum F_z = 0 = -1600 + 0.836T_{AB} + 0.91T_{AC} + 0.761T_{AD} \Rightarrow III$$

Hence, by solve I, II and III:

$$\therefore T_{AB} = 739.0352 \text{ N}$$

$$\therefore T_{AC} = 932.942 \text{ N}$$

$$\therefore T_{AD} = 175.022 \text{ N}$$

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